

PRE-ENGINEERING PATHWAY

Courses

Below you will find a list of our current or recent offerings. See the course catalog for descriptions and updated information.

CHEMISTRY

COMPUTER SCIENCE

MATHEMATICS

PHYSICS

- CALCULUS I (52-154). Functions and graphs; derivatives, applications of differentiation. Exponential, logarithmic and trigonometric functions, integration, applications of integration. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Prerequisite: Mastery of high school level precalculus (algebra, trigonometry, exponential and logarithmic functions). (Each semester) (NS)
- CALCULUS II (52-254). Numerical integration, methods of integration, applications of the definite integral, improper integrals, sequences and series, Taylor’s Formula and approximation, polar coordinates. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Prerequisite: Mathematics 52-154. (Each semester) (NS)
- CALCULUS III (52-353). A course in multivariable calculus. Vectors, vector functions, and curves. Functions of several variables, partial differentiation, multiple integration, applications of partial differentiation and of multiple integrals. Vector calculus, line integrals, Green’s Theorem, surface integrals. Prerequisite: Mathematics 52-254. (Each semester) (NS)
- LINEAR ALGEBRA (52-673). Linear equations and matrices, vector spaces, linear mappings, determinants, quadratic forms, vector products, groups of symmetries. Prerequisite: Mathematics 52-254 or permission of instructor. (Each semester) (NS)
- ELEMENTARY DIFFERENTIAL EQUATIONS (52-753). Topics include first order differential equations, separable equations, exact equations, linear differential equations of order $n > 1$, homogeneous equations with constant coefficients, nonhomogeneous equations, the method of undetermined coefficients, variation of parameters, power series solutions, and an introduction to the Laplace Transform. Prerequisite: Mathematics 52-353, or permission of instructor. (Fall) (NS)